

Physics Origins 03: From Kinetic Energy Integration to Probability Amplitude Squared – An Intuitive Chain

Core Logic Flow: Intuition → Modeling → Mathematical Derivation →
Experimental Validation

This document builds the entire logical chain from the ground up, starting with everyday intuition (apples occupying volume), then moving to mathematical modeling, closing with calculus-based derivation, and finally validating with experimental facts. No reverse engineering, no circular reasoning.

Part 1: Everyday Intuition – Density Is "1 / Volume per Unit"

1.1 A Daily Scene

Suppose 100 apples are evenly stacked in a 20 m³ room.

Question: How dense are the apples in this room?

Intuitive answer: $100 \div 20 = 5$ apples/m³.

1.2 Another Way to Say It (This Is the Seed of the Whole Logic)

You can also calculate it like this:

- $20 \text{ m}^3 \div 100 \text{ apples} = 0.2 \text{ m}^3$ per apple (each apple occupies 0.2 m³ on average).
- Then $1 \div 0.2 = 5$.

This "5" is the number of apples per cubic meter – that is, the density.

1.3 Abstract This Intuition into a Formula

Density = 1 / (Average volume occupied by a single unit)

Or equivalently:

Density = Total count / Total volume

Part 2: From Apples to Quantum Waves (Modeling)

2.1 Quantum Mechanics Says: Particles Are "Waves"

In the microscopic world, an electron is not a hard little ball, but a "probability cloud" spread out in space.

The more this cloud gathers in a certain place, the higher the chance you will find the particle there.

2.2 How Do We Describe the "Degree of Gathering" of This Cloud?

In classical physics (water waves, sound waves, light waves), the "intensity" of a wave is proportional to the **square of the amplitude (A^2)**.

Therefore, physicists define a mathematical function for the electron, called the wavefunction Ψ , and stipulate:

$$\text{Wave intensity} = |\Psi|^2$$

2.3 Transfer the Apple Intuition Over

- Apple density = 5 apples/m³ → means "how many apples are stacked per cubic meter".
- $|\Psi|^2 = 5$ (with unit 1/m³) → means "how much probability share of the particle is stacked per cubic meter".

They are exactly the same thing.

$|\Psi|^2$ is the **probability density** of the particle in space – the amount of "probability of appearance" per unit volume.

Part 3: Calculus Derivation of the Kinetic Energy Formula (The Root of the Math)

Many people know kinetic energy is $\frac{1}{2}mv^2$, but don't know where that "square" comes from. We start from the most primitive "work = force × displacement" and derive it using calculus. You'll see that this "square" is a natural product of integral operations, not an arbitrary human stipulation.

3.1 Newton's Second Law (Differential Form)

Force = rate of change of momentum. When mass is constant:

$$F = m \times (dv / dt)$$

3.2 Differential Definition of Work

The infinitesimal work = force × infinitesimal displacement:

$$dW = F \times ds$$

3.3 Replace Displacement with Velocity

Since $v = ds / dt$, we have $ds = v \times dt$. Substitute:

$$dW = F \times v \times dt$$

3.4 Substitute Newton's Definition ($F = m \times dv/dt$)

$$dW = m \times (dv/dt) \times v \times dt$$

The dt cancels out:

$$dW = m \times v \times dv$$

3.5 Integrate Both Sides (from velocity 0 to v)

$$W = \int_0^v m \times v \times dv = m \times \int_0^v v \times dv = \frac{1}{2} m v^2$$

This derivation reveals: the "square" comes from the integral $\int v \, dv = \frac{1}{2}v^2$.

As long as force = rate of change of momentum, the integral of work will inevitably produce v^2 .

Part 4: Using the "Apple Intuition" to Completely Resolve Dimensions (Core Logic)

4.1 Recall Your Original Statement (This Is the Truest Intuition)

"100 apples occupy 20 m³, so one apple occupies 0.2 m³, then 1/0.2 = 5, which is 5 apples per cubic meter – that is density. So energy density is how much energy (N particles) occupies unit space. Total energy is M particles. Then N/M represents the probability of a particle appearing in one cubic meter, similar to the probability of a point falling into a volume."

This statement has already **completely walked through** the logic of probability density in quantum mechanics. We just need to translate it into physical language.

4.2 Replace "Apples" with "Probability Cloud"

- A small volume dV contains N shares of "probability share."
- The entire space has a total of M shares, and since total probability = 1, $M = 1$.
- So the probability of finding the particle in that volume = $N/M = N = \text{density} \times \text{volume}$.

Density is $|\Psi|^2$, volume is dV , so:

$$\text{Probability} = |\Psi|^2 \times dV$$

4.3 Dimensions Naturally Cancel

- $|\Psi|^2$ has unit $1/m^3$ (density – same as "5 apples/ m^3 ").
- dV has unit m^3 (volume).
- Multiply $\rightarrow$ units cancel, leaving a pure number (probability).

This is why $|\Psi|^2$ is called "probability density" – because it must be multiplied by a volume to yield a true probability.

4.4 This Resonates with the Intuition of Maximum Likelihood Estimation (MLE)

"N/M represents the probability of a particle appearing in one cubic meter" is essentially using a local proportion to estimate the global probability, in the same spirit as MLE.

Part 5: Why Must It Be the 2nd Power, Not the 4th? (Validation)

5.1 Classical Root

Classical wave equations are **linear**. In linear waves, velocity $v \propto$ amplitude A , so energy $\propto v^2 \propto A^2$.

5.2 What If the World Were 4th Power?

Probability = $|\Psi|^4 \rightarrow$ amplitude doubles, count becomes $16\times$, but energy only $4\times \rightarrow$ energy non-conservation or spurious cross-terms.

5.3 Conclusion

The "2nd power" is the unique exponent locked in by linear wave equations + energy conservation.

Part 6: Ultimate Clarification of the Role of "Force = Rate of Change of Momentum"

Core Question: Can "force = rate of change of momentum" derive $P = |\Psi|^2$ from the ground up?

Answer: No.

It is a **post-hoc validator**, not a **pre-hoc derivator**.

The flow: experiment $\rightarrow$ induce $P = |\Psi|^2 \rightarrow$ construct dynamical equations $\rightarrow$ macroscopic limit recovers $F = ma$.

So it only proves that quantum mechanics does not conflict with Newtonian mechanics, not that it originates from them.

Part 7: Complete Logical Chain (Overview)

Step	Content	Nature
1	Apples occupying volume $\rightarrow$ density = 1/unit volume	Everyday Intuition
2	Electron diffraction $\rightarrow$ fringe distribution = wave intensity distribution	Experimental Fact
3	Born's Rule: probability density =	Ψ
4	Kinetic integration: $\int m \cdot v \cdot dv = \frac{1}{2}mv^2$	Mathematical Derivation
5	Probability = density $\times$ volume =	Ψ
6	Macroscopic limit reduces to Newtonian mechanics	Consistency Check

Part 8: The Final Answer

"Why is probability the squared modulus?"

Because the shape of experimental data (electron diffraction fringes) forces physicists to use the "square of the amplitude" to describe the appearance pattern of electrons. This is an axiom induced from experiment, not a derivation from Newton's laws.

"Energy and probability have different dimensions – how are they related?"

Probability is density $\times$ volume. $|\Psi|^2$ has unit $1/m^3$, multiplied by dV (m^3) gives a pure number, exactly the same logic as the apples.

Appendix: Key Formulas in Plain Text

- Kinetic derivation: $W = \int_0^v m \cdot v \cdot dv = \frac{1}{2} m v^2$
- Probability density: $|\Psi|^2$ (unit $1/m^3$)
- Probability in a volume: $P = |\Psi|^2 \times dV$ (dimensionless)
- Normalization: $\int |\Psi|^2 dV = 1$
- Apple analogy: Density = total count / total volume = 1 / unit volume